

Mayank Soni

Email | [GitHub](#) | [Portfolio](#)

EDUCATION

- **University Of Hyderabad**
Integrated B.Sc. + M.Sc. in Physics,
2023 - Expected 2028
-

RESEARCH EXPERIENCE

Undergraduate Researcher

University of Hyderabad
Advisor: Prof. Nirmal K. Viswanathan
June 2025 - Present

- Investigated three-dimensional polarization singularities in electromagnetic vector fields.
- Developed computational methods for simulation and analysis of optical speckle fields.
- Created numerical tools for visualization and tracking of optical singular structures.
- Contributed to preparation of a manuscript on polarization singularities and optical speckle fields.

Summer Research Intern

Krittika Summer Projects, IIT Bombay
May 2024 - August 2024

- Analyzed observational data from SN 2023ixf using Python.
 - Constructed multi-band light curves from astronomical observations.
 - Estimated physical parameters of the supernova through numerical analysis.
-

SCIENTIFIC SOFTWARE

VectorWaves

December 2025 - June 2026

- Developed an open-source Python package for construction and analysis of electromagnetic fields using plane-wave expansions ([GitHub](#)).
 - Implemented GPU accelerated field evaluation, visualization, and singularity-analysis workflows.
 - Published package on [PyPI](#) and created accompanying [documentation](#).
-

PRESENTATIONS & MANUSCRIPT

- **Oral Presentation:** “Three-Dimensional Polarization Structure: From Paraxial to Full-Vector Singularities”, SCOPOSIS-2025, Physical Research Laboratory, Ahmedabad.
 - **Poster Presentation:** “Three-Dimensional Analysis of Polarisation Singularities, Paraxial to Isotropic Regimes”, 100 Years of Quantum Mechanics, University of Hyderabad, Hyderabad.
 - **Manuscript in Preparation:** “Dimensional Crossover in Gaussian Speckles and its Topological Consequences”, Mayank Soni, Nirmal K. Viswanathan.
-

TECHNICAL SKILLS

- Programming: Python, Julia
- Scientific Computing: NumPy, SciPy, Matplotlib
- Tools: Git, LaTeX